

Dropout Multilayer Perceptron

Notes

Output Created		27-APR-2026 12:01:42
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	600
Missing Value Handling	Definition of Missing	User- and system-missing values are treated as missing.
	Cases Used	Statistics are based on cases with valid data for all variables used by the procedure.
Weight Handling		not applicable
Syntax		MLP Dropped_Out_Dummy (MLEVEL=N) WITH HS_GPA Unit_Enrolled Seminar_D Commuter_Dummy /RESCALE COVARIATE=STANDARDIZED /PARTITION TRAINING=7 TESTING=3 HOLDOUT=0 /ARCHITECTURE AUTOMATIC=YES (MINUNITS=1

		MAXUNITS=50) /CRITERIA TRAINING=BATCH OPTIMIZATION=SCALE DCONJUGATE LAMBDAINITIAL=0.000 0005 SIGMAINITIAL=0.00005 INTERVALCENTER=0 INTERVALOFFSET=0.5 MEMSIZE=1000 /PRINT CPS NETWORKINFO SUMMARY CLASSIFICATION /PLOT NETWORK ROC /STOPPINGRULES ERRORSTEPS= 1 (DATA=AUTO) TRAININGTIMER=ON (MAXTIME=15) MAXEPOCHS=AUTO ERRORCHANGE=1.0E- 4 ERRORRATIO=0.001 /MISSING USERMISSING=EXCLU DE .
Resources	Processor Time	00:00:03.30
	Elapsed Time	00:00:00.85

[DataSet1]

Case Processing Summary

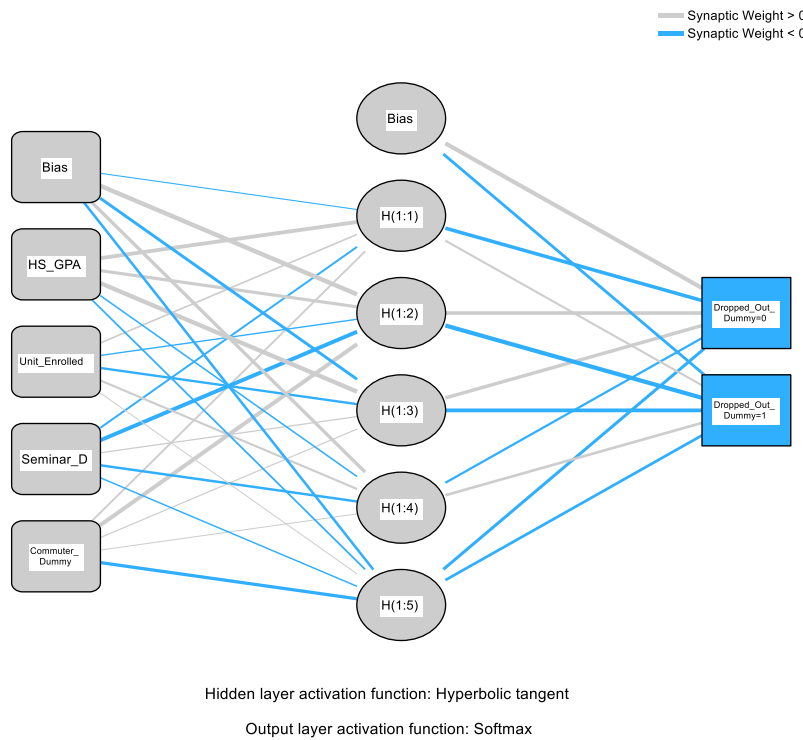
	N	Percent
Sample Training	403	67.2%

Testing	197	32.8%
Valid	600	100.0%
Excluded	0	
Total	600	

Network Information

Input Layer	Covariates	1	HS_GPA
		2	Unit_Enrolled
		3	Seminar_D
		4	Commuter_Dummy
	Number of Units ^a		4
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		5
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	Dropped_Out_Dummy
	Number of Units		2
	Activation Function		Softmax
	Error Function		Cross-entropy

a. Excluding the bias unit



Model Summary

Training	Cross Entropy Error	171.319
	Percent Incorrect Predictions	19.1%
	Stopping Rule Used	1 consecutive step(s) with no decrease in error ^a
	Training Time	0:00:00.04
Testing	Cross Entropy Error	82.628
	Percent Incorrect	20.8%

Predictions

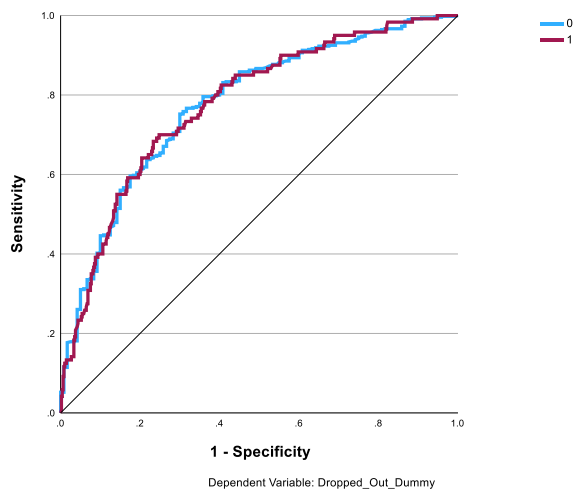
Dependent Variable: Dropped_Out_Dummy

a. Error computations are based on the testing sample.

Classification

Sample	Observed	Predicted		Percent Correct
		0	1	
Training	0	316	12	96.3%
	1	65	10	13.3%
	Overall Percent	94.5%	5.5%	80.9%
Testing	0	149	3	98.0%
	1	38	7	15.6%
	Overall Percent	94.9%	5.1%	79.2%

Dependent Variable: Dropped_Out_Dummy



Area Under the Curve

		Area
Dropped_Out_Dum	0	.777
my	1	.777

Dropout Logistic Regression

Notes

Output Created		22-SEP-2026 12:14:34
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	600
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing
Syntax	LOGISTIC REGRESSION VARIABLES Dropped_Out_Dummy /METHOD=ENTER HS_GPA Unit_Enrolled Commuter_Dummy Seminar_D /CLASSPLOT /PRINT=GOODFIT CI(90) /CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).	
Resources	Processor Time	00:00:00.02

Elapsed Time	00:00:00.02
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Case Processing Summary

Unweighted Cases ^a		N	Percent
Selected Cases	Included in Analysis	600	100.0
	Missing Cases	0	.0
	Total	600	100.0
Unselected Cases		0	.0
Total		600	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
0	0
1	1

Block 0: Beginning Block

Classification Table^{a,b}

Observed			Predicted		Percentage Correct
			Dropped_Out_Dum my	0	
Step 0	Dropped_Out_Dum	0	480	0	100.0
	my	1	120	0	.0
	Overall Percentage				80.0

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

	B	S.E.	Wald	df	Sig.	Exp(B)
Step 0 Constant	-1.386	.102	184.494	1	<.001	.250

Variables not in the Equation

	Score	df	Sig.
Step 0 Variables HS_GPA	42.979	1	<.001
Unit_Enrolled	14.704	1	<.001
Commuter_Dummy	9.435	1	.002
Seminar_D	11.076	1	<.001
Overall Statistics	75.397	4	<.001

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

	Chi-square	df	Sig.
Step 1 Step	82.180	4	<.001
Block	82.180	4	<.001
Model	82.180	4	<.001

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	518.303 ^a	.128	.202

a. Estimation terminated at iteration number 5 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
1	12.933	8	.114

Contingency Table for Hosmer and Lemeshow Test

		Dropped_Out_Dummy = 0		Dropped_Out_Dummy = 1		Total
		Observed	Expected	Observed	Expected	
Step 1	1	60	58.270	0	1.730	60
	2	53	56.590	7	3.410	60
	3	52	54.972	8	5.028	60
	4	55	53.335	5	6.665	60
	5	54	51.360	6	8.640	60
	6	51	49.236	9	10.764	60
	7	50	46.680	10	13.320	60
	8	41	42.983	19	17.017	60
	9	33	37.898	27	22.102	60
	10	31	28.677	29	31.323	60

Classification Table^a

			Predicted		Percentage Correct
			Dropped_Out_Dum my		
Step 1	Observed		0	1	
	Dropped_Out_Dum my	0	468	12	97.5
		1	104	16	13.3
Overall Percentage					80.7

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.	Exp(B)
Step 1 ^a	HS_GPA	-1.621	.275	34.741	1	<.001	.198
	Unit_Enrolled	.147	.034	19.062	1	<.001	1.158

Commuter_Dummy	-.800	.268	8.912	1	.003	.449
Seminar_D	.673	.226	8.831	1	.003	1.960
Constant	1.206	.834	2.091	1	.148	3.340

Variables in the Equation

		90% C.I. for EXP(B)	
		Lower	Upper
Step 1 ^a	HS_GPA	.126	.311
	Unit_Enrolled	1.096	1.224
	Commuter_Dummy	.289	.698
	Seminar_D	1.351	2.845
	Constant		

a. Variable(s) entered on step 1: HS_GPA, Unit_Enrolled, Commuter_Dummy, Seminar_D.

Step number: 1

Observed Groups and Predicted Probabilities

32	+	1
+		
I	I	1
I	I	1
I	I	1
F	I	0
I		

```

R      24 +      10 0 0
+
E      I      10 010
I
Q      I  0000 010  101
I
U      I  00001000 1000 11  0      1
I
E      16 +  00001000 0000 101 00      0
+
N      I 00000000000000001000 00      0
I
C      I 00000000000000001000 001  0 0      1
I
Y      I 000000000000000000001001 10 0 1  11      1
I
      8 + 000000000000000000000000100 0 1 111  1 10 1      1
+
      I 0000000000000000000000000000 010 111 111000 1      0      1
I
      I 00000000000000000000000000001001 00000000 11 0 1      1
I
      I 0000000000000000000000000000000000000000000000000000000 0010000
0001100 0 00101 1  110      I
Predicted -----+-----+-----+-----+-----+-----
-----+-----+-----+-----+-----
Prob:   0      .1      .2      .3      .4      .5
.6      .7      .8      .9      1
Group:
0000000000000000000000000000000000000000000000000000000011111111111111
11111111111111111111111111111111111111111111111111111111111111111111

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Predicted Probability is of Membership for 1
 The Cut Value is .50
 Symbols: 0 - 0
 1 - 1
 Each Symbol Represents 2 Cases.

Seminar Logistic Regression

Notes

Output Created		22-SEP-2026 12:51:34
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	600
Missing Value Handling	Definition of Missing	User-defined missing values are treated as missing
Syntax		LOGISTIC REGRESSION VARIABLES Seminar_D /METHOD=ENTER HS_GPA Unit_Enrolled Commuter_Dummy Age Gender_Dummy Registration_Status_Du mmy /CLASSPLOT /PRINT=GOODFIT CI(90) /CRITERIA=PIN(0.05) POUT(0.10) ITERATE(20) CUT(0.5).
Resources	Processor Time	00:00:00.00
	Elapsed Time	00:00:00.01

Case Processing Summary

Unweighted Cases ^a	N	Percent
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Selected Cases	Included in Analysis	600	100.0
	Missing Cases	0	.0
	Total	600	100.0
Unselected Cases		0	.0
Total		600	100.0

a. If weight is in effect, see classification table for the total number of cases.

Dependent Variable Encoding

Original Value	Internal Value
0	0
1	1

Block 0: Beginning Block

Classification Table^{a,b}

Observed		Predicted		Percentage Correct
		Seminar_D 0	1	
Step 0	Seminar_D 0	374	0	100.0
	1	226	0	.0
	Overall Percentage			62.3

a. Constant is included in the model.

b. The cut value is .500

Variables in the Equation

B	S.E.	Wald	df	Sig.	Exp(B)
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Step 0	Constant	-.504	.084	35.744	1	<.001	.604
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Variables not in the Equation

		Score	df	Sig.
Step 0	Variables			
	HS_GPA	10.961	1	<.001
	Unit_Enrolled	.038	1	.845
	Commuter_Dummy	19.584	1	<.001
	Age	20.245	1	<.001
	Gender_Dummy	2.012	1	.156
	Registration_Status_Dummy	.522	1	.470
	Overall Statistics	57.038	6	<.001

Block 1: Method = Enter

Omnibus Tests of Model Coefficients

		Chi-square	df	Sig.
Step 1	Step	58.223	6	<.001
	Block	58.223	6	<.001
	Model	58.223	6	<.001

Model Summary

Step	-2 Log likelihood	Cox & Snell R Square	Nagelkerke R Square
1	736.668 ^a	.092	.126

a. Estimation terminated at iteration number 4 because parameter estimates changed by less than .001.

Hosmer and Lemeshow Test

Step	Chi-square	df	Sig.
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1	3.419	8	.905
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Contingency Table for Hosmer and Lemeshow Test

		Seminar_D = 0		Seminar_D = 1		Total
		Observed	Expected	Observed	Expected	
Step 1	1	50	48.942	10	11.058	60
	2	46	46.136	14	13.864	60
	3	46	44.202	14	15.798	60
	4	42	42.320	18	17.680	60
	5	42	40.368	18	19.632	60
	6	32	38.094	28	21.906	60
	7	36	35.384	24	24.616	60
	8	33	32.112	27	27.888	60
	9	28	28.255	32	31.745	60
	10	19	18.187	41	41.813	60

Classification Table^a

		Predicted		
		Seminar_D		Percentage Correct
Observed		0	1	
Step 1	Seminar_D 0	330	44	88.2
	1	156	70	31.0
	Overall Percentage			66.7

a. The cut value is .500

Variables in the Equation

		B	S.E.	Wald	df	Sig.
Step 1 ^a	HS_GPA	-.783	.197	15.825	1	<.001
	Unit_Enrolled	-.012	.027	.190	1	.663
	Commuter_Dummy	.910	.190	22.901	1	<.001
	Age	.085	.019	18.852	1	<.001
	Gender_Dummy	.290	.178	2.651	1	.104
	Registration_Status_Du	.192	.214	.798	1	.372

mmy					
Constant	-.500	.737	.459	1	.498

Variables in the Equation

			90% C.I. for EXP(B)	
			Lower	Upper
Step 1 ^a	HS_GPA	.457	.331	.632
	Unit_Enrolled	.988	.946	1.033
	Commuter_Dummy	2.485	1.818	3.398
	Age	1.088	1.054	1.124
	Gender_Dummy	1.336	.997	1.790
	Registration_Status_Dummy	1.211	.851	1.723
	Constant	.607		

a. Variable(s) entered on step 1: HS_GPA, Unit_Enrolled, Commuter_Dummy, Age, Gender_Dummy , Registration_Status_Dummy.

Step number: 1

Observed Groups and Predicted Probabilities

32 +
+
I
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F I

Seminar Multilayer Perceptron

Notes

Output Created		22-SEP-2026 12:57:14
Comments		
Input	Active Dataset	DataSet1
	Filter	<none>
	Weight	<none>
	Split File	<none>
	N of Rows in Working Data File	600
Missing Value Handling	Definition of Missing	User- and system-missing values are treated as missing.
	Cases Used	Statistics are based on cases with valid data for all variables used by the procedure.
Weight Handling		not applicable
Syntax		MLP Seminar_D (MLEVEL=N) WITH HS_GPA Unit_Enrolled Commuter_Dummy Age Gender_Dummy Registration_Status_Dummy /RESCALE COVARIATE=STANDARDIZED /PARTITION TRAINING=7 TESTING=3 HOLDOUT=0 /ARCHITECTURE AUTOMATIC=YES

		(MINUNITS=1 MAXUNITS=50) /CRITERIA TRAINING=BATCH OPTIMIZATION=SCALE DCONJUGATE LAMBDAINITIAL=0.000 0005 SIGMAINITIAL=0.00005 INTERVALCENTER=0 INTERVALOFFSET=0.5 MEMSIZE=1000 /PRINT CPS NETWORKINFO SUMMARY CLASSIFICATION /PLOT NETWORK ROC /STOPPINGRULES ERRORSTEPS= 1 (DATA=AUTO) TRAININGTIMER=ON (MAXTIME=15) MAXEPOCHS=AUTO ERRORCHANGE=1.0E- 4 ERRORRATIO=0.001 /MISSING USERMISSING=EXCLU DE .
Resources	Processor Time	00:00:00.64
	Elapsed Time	00:00:00.48

Case Processing Summary

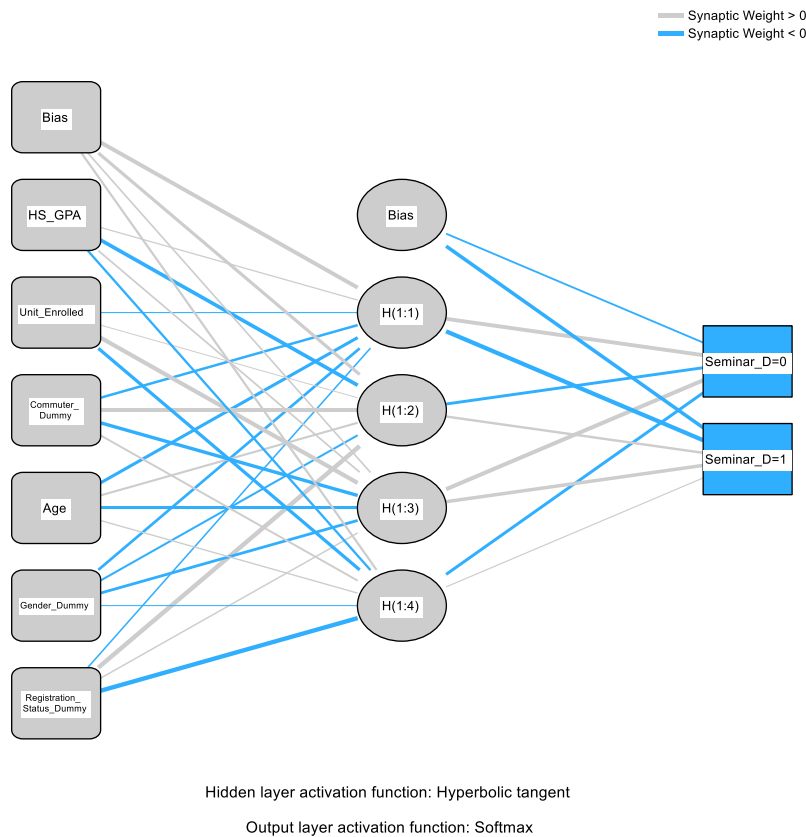
		N	Percent
Sample	Training	429	71.5%
	Testing	171	28.5%
Valid		600	100.0%
Excluded		0	

Total	600
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Network Information

Input Layer	Covariates	1	HS_GPA
		2	Unit_Enrolled
		3	Commuter_Dummy
		4	Age
		5	Gender_Dummy
		6	Registration_Status_Dummy
	Number of Units ^a		6
	Rescaling Method for Covariates		Standardized
Hidden Layer(s)	Number of Hidden Layers		1
	Number of Units in Hidden Layer 1 ^a		4
	Activation Function		Hyperbolic tangent
Output Layer	Dependent Variables	1	Seminar_D
	Number of Units		2
	Activation Function		Softmax
	Error Function		Cross-entropy

a. Excluding the bias unit



Model Summary

Training	Cross Entropy Error	274.538
	Percent Incorrect Predictions	37.5%
	Stopping Rule Used	1 consecutive step(s) with no decrease in error ^a

	Training Time	0:00:00.05
Testing	Cross Entropy Error	95.718
	Percent Incorrect Predictions	24.0%

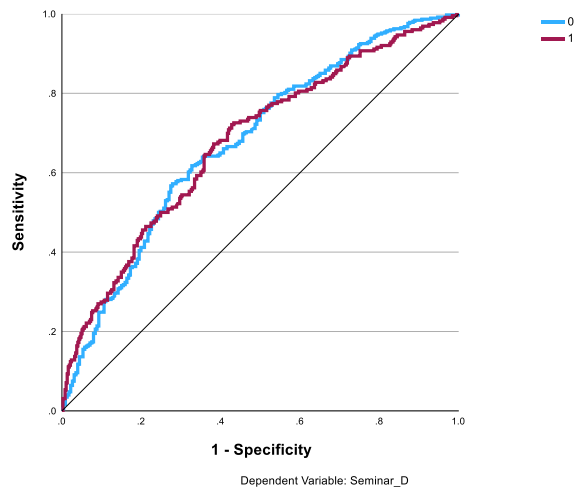
Dependent Variable: Seminar_D

a. Error computations are based on the testing sample.

Classification

Sample	Observed	Predicted		Percent Correct
		0	1	
Training	0	227	33	87.3%
	1	128	41	24.3%
	Overall Percent	82.8%	17.2%	62.5%
Testing	0	109	5	95.6%
	1	36	21	36.8%
	Overall Percent	84.8%	15.2%	76.0%

Dependent Variable: Seminar_D



Area Under the Curve

		Area
Seminar_D	0	.677
	1	.677